

Regression Analysis (II)

Project 1.

Due October 28, 2019

You may use any statistical packages like R, minitab, spss, sas, etc.

1. Make your own dataset based on data in Example 6.5 (p. 251). Let $X_1 \leftarrow X_1 + \epsilon$, $X_2 \leftarrow X_2 + \epsilon$, $Y \leftarrow Y + \epsilon$, where $\epsilon \sim N(0, 1)$. Do the response surface analysis with contour plot.
2. Make your own dataset based on data in Example 6.6 (p. 254). Let $X \leftarrow X_1 + \epsilon$, $Y \leftarrow Y + \epsilon$, where $\epsilon \sim N(0, 0.1^2)$. Compute the WLSE using the same method as the one in text.
3. Make your own dataset based on data in Example 6.7 (p. 259). Let $Y \leftarrow Y + \epsilon$, where $\epsilon \sim N(0, 1)$.
 - (1) Fit the data to the multiple linear regression model.
 - (2) Fit the data to the Box-Cox transformation model.
 - (3) Compare two models in (1) and (2) by using the $Q - Q$ plot of residuals in each model.